

Generation Interconnection System Impact Study Report

For Queue Project Q12-0153

J R Whiting 100 kV

April 01, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0153. The Interconnection Customer proposes a 541.0 MW natural gas combined cycle facility interconnecting at the J R Whiting 100 kV substation in the Michigan area. The request was received on September 01, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 108 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,838,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Unc Qf / Boyles 100 KV Line
Contingency: J R Whiting 230/1kV Autotransformer
Loading: 180.1% (Rating: 107 MVA)
DFAX: 52.1%
Pre-project loading: 83.3%
2. Facility: Black Hills Power / Boyles 100 KV
Contingency: J R Whiting 230 kV/1 kV TX
Loading: 173.3% (Rating: 121 MVA)
DFAX: 52.1%
Pre-project loading: 60.0%
3. Facility: S Woodrow / Black Hills Power 100 KV
Contingency: J R Whiting 230-1 kV Auto
Loading: 172.0% (Rating: 130 MVA)
DFAX: 52.1%
Pre-project loading: 45.1%
4. Facility: Carrollton-Upfield / Unc Qf 100 KV
Contingency: J R Whiting Substation 230 kV/1 kV Auto
Loading: 161.1% (Rating: 115 MVA)
DFAX: 52.1%
Pre-project loading: 83.7%
5. Facility: Deer Park / Koch Oil 100 KV
Contingency: J R Whiting 230-1 kV TX
Loading: 146.0% (Rating: 122 MVA)
DFAX: 52.1%
Pre-project loading: 85.6%
6. Facility: Carrollton-Upfield / Koch Oil 100 KV Line
Contingency: J R Whiting 230/1kV Transformer Bank
Loading: 127.2% (Rating: 133 MVA)

- DFAX: 52.1%
Pre-project loading: 84.1%
7. Facility: Doyle Energy Facility / Toll Rd 100 KV
Contingency: Fermi 500-230 kV Auto
Loading: 115.2% (Rating: 138 MVA)
DFAX: 5.8%
Pre-project loading: 92.3%
8. Facility: Toll Rd / Bonita Springs 100 KV
Contingency: Fermi 500/230 kV Transformer
Loading: 111.9% (Rating: 142 MVA)
DFAX: 5.8%
Pre-project loading: 89.7%
9. Facility: Doyle Energy Facility / Lewisville Switch 100 KV line
Contingency: Fermi 500-230 kV Transformer
Loading: 106.4% (Rating: 143 MVA)
DFAX: 5.8%
Pre-project loading: 84.3%
10. Facility: Bonita Springs / Midway M.P 100 KV
Contingency: Fermi 500-230 kV Transformer Bank
Loading: 103.7% (Rating: 356 MVA)
DFAX: 8.6%
Pre-project loading: 90.6%
11. Facility: Post Rock / Newport Tie 230 KV
Contingency: Trego / Beaver 500 KV line
Loading: 102.8% (Rating: 543 MVA)
DFAX: 14.2%
Pre-project loading: 88.6%
12. Facility: Myrtle Point / Post Rock Substation 230 KV
Contingency: Trego / Beaver 500 KV
Loading: 102.5% (Rating: 621 MVA)
DFAX: 13.6%
Pre-project loading: 90.6%
13. Facility: Corbell / Deer Park 100 KV
Contingency: J R Whiting 230/1kV XFMR
Loading: 102.4% (Rating: 152 MVA)
DFAX: 52.1%
Pre-project loading: 83.5%
14. Facility: Joso / Onancock Diesel Generation Facility 230 KV
Contingency: Beaver / Eufaula 500 KV
Loading: 101.7% (Rating: 537 MVA)
DFAX: 17.3%
Pre-project loading: 84.3%
15. Facility: Beaver 230/1kV Auto
Contingency: Eufaula / Beaver 500 KV
Loading: 101.0% (Rating: 1,021 MVA)
DFAX: 14.5%
Pre-project loading: 93.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Boyles / Unc Qf 100 KV	Rebuild line with higher-capacity conductor	\$3,176,000	100% (\$3,176,000)
Black Hills Power / Boyles 100 KV	Rebuild line with higher-capacity conductor	\$4,689,000	100% (\$4,689,000)
S Woodrow / Black Hills Power 100 KV	Rebuild line with higher-capacity conductor	\$2,178,000	100% (\$2,178,000)
Carrollton-Upfield / Unc Qf 100 KV line	Rebuild line with higher-capacity conductor	\$6,567,000	100% (\$6,567,000)
Deer Park / Koch Oil 100 KV	Rebuild line with higher-capacity conductor	\$2,689,000	100% (\$2,689,000)
Carrollton-Upfield Substation / Koch Oil 100 KV	Rebuild line with higher-capacity conductor	\$1,525,000	100% (\$1,525,000)
Doyle Energy Facility / Toll Rd 100 KV	Rebuild line with higher-capacity conductor	\$3,760,000	100% (\$3,760,000)
Bonita Springs / Toll Road Substation 100 KV	Rebuild line with higher-capacity conductor	\$5,015,000	100% (\$5,015,000)
Lewisville Switch / Doyle Energy Facility Substation 100 KV	Reconductor line	\$1,425,000	100% (\$1,425,000)
Bonita Springs / Midway M.P 100 KV	Reconductor line	\$1,687,000	100% (\$1,687,000)
Post Rock / Newport Tie 230 KV	Reconductor line	\$11,389,000	50% (\$5,648,000)
Post Rock / Myrtle Pt 230 KV line	Reconductor line	\$8,848,000	50% (\$4,388,000)
Corbell / Deer Park 100 KV	Reconductor line	\$262,000	100% (\$262,000)
Joso / Onancock Diesel Generation Facility 230 KV	Reconductor line	\$6,430,000	56% (\$3,583,000)
Beaer 230/1kV Auto	Replace transformer with higher-rated unit	\$8,534,000	50% (\$4,246,000)
	Total allocated to project		\$50,838,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.